Taiwan Semiconductor Manufacturing Company Ltd
(NYSE: TSM)

Sector: TMT
7 October 2023

CONTENTS

EXECUTIVE SUMMARY

Ticker
NYSE: TSM

Stock Price
$90.48

Market Cap.
$469.99B

Target Price (Nov 24E)
$103.00

UPSIDE: 14%

COMPANY OVERVIEW

Business Overview

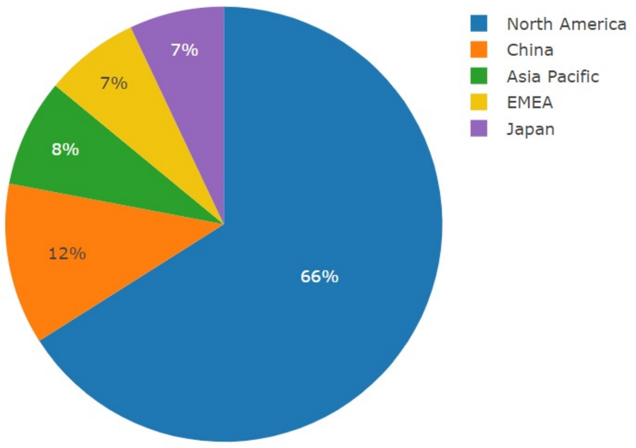
- World's largest independent pure-play semiconductor foundry based in Taiwan, with a **56.4%** market share
- Customers consist of leaders in their respective fields such as Apple, AMD, Nvidia, Qualcomm
- TSMC produces **90%** of the advanced chips used in AI in the industry

Key Financials

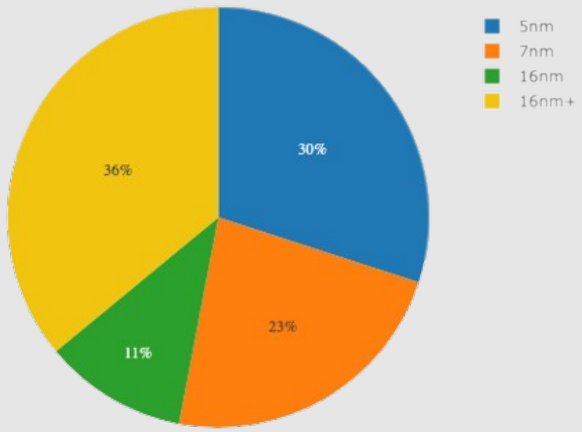
- **Market Cap:** $469.99B
- **ROA (TTM):** 19.84%
- **Operating margin (TTM):** 47.96%
- **P/E:** 15.0x

MSCI ESG Rating: AAAA

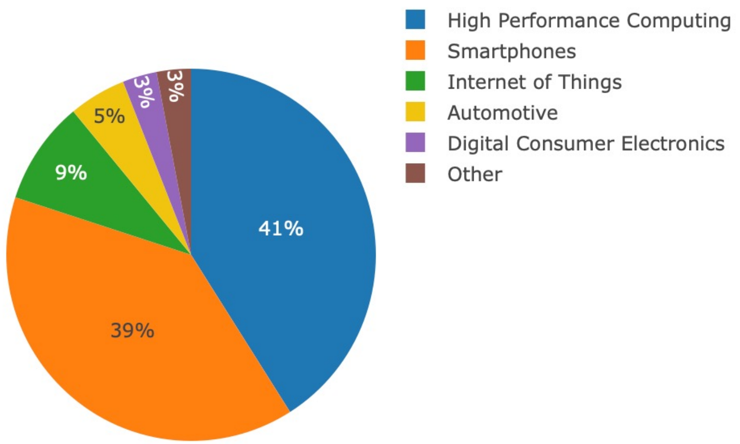
Net Revenue by Geography (2Q23)



Wafer Revenue by Technology (2Q23)



Net Revenue by Platform (2Q23)



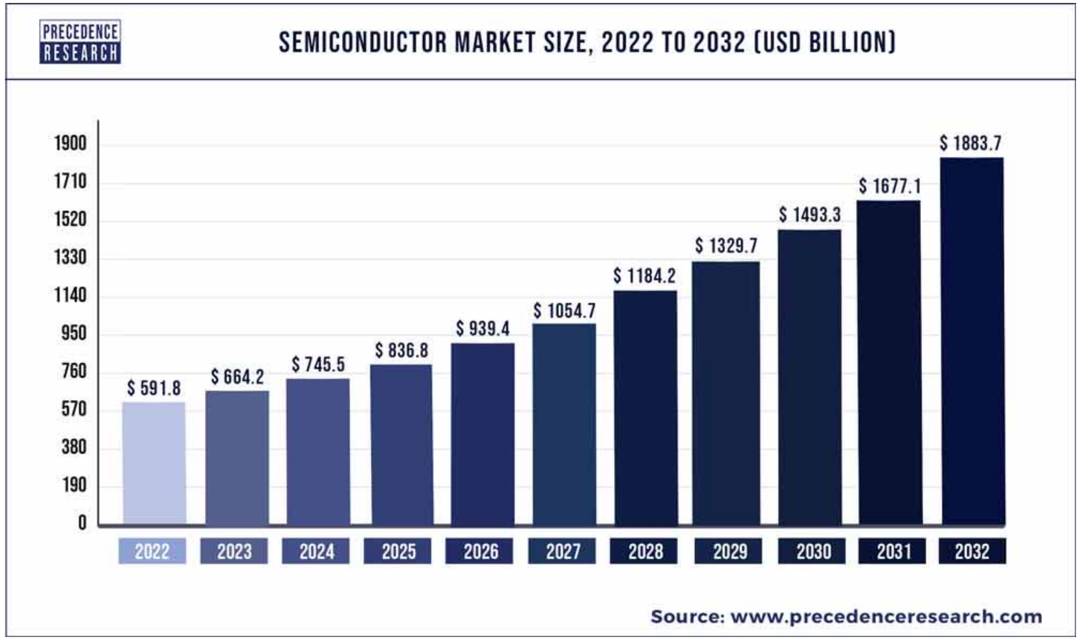
INDUSTRY OVERVIEW

Semiconductor industry expected to grow consistently…

12.2% CAGR
2022-2029E

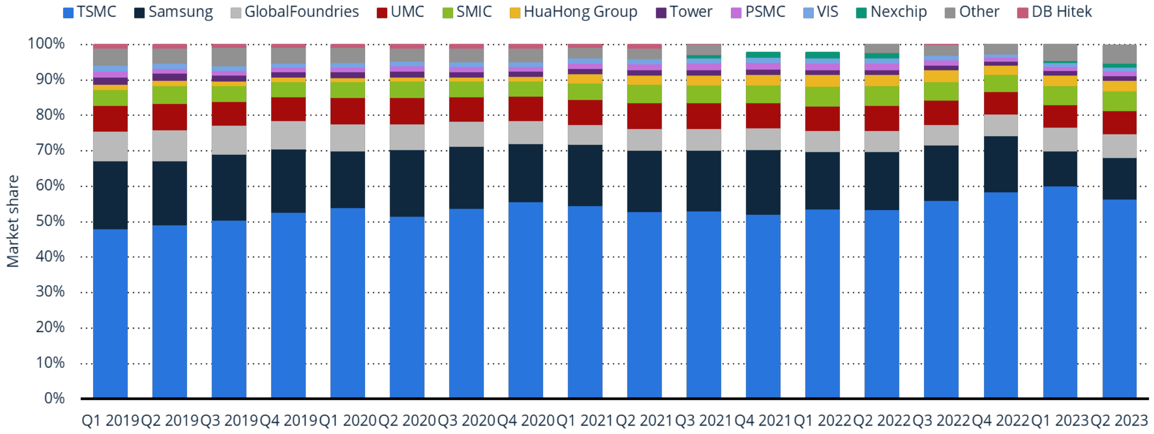
Driven by the demand for digital solutions such as:

- Artificial Intelligence (AIs)
- Data Storage
- Wireless Communications
- Electric Vehicles (EVs)
- IoTs



…with TSMC as a leader in the FOUNDRY market

- TSMC is the biggest global semiconductors foundry player >50% market share
- Very high barriers to entry



What is foundry?

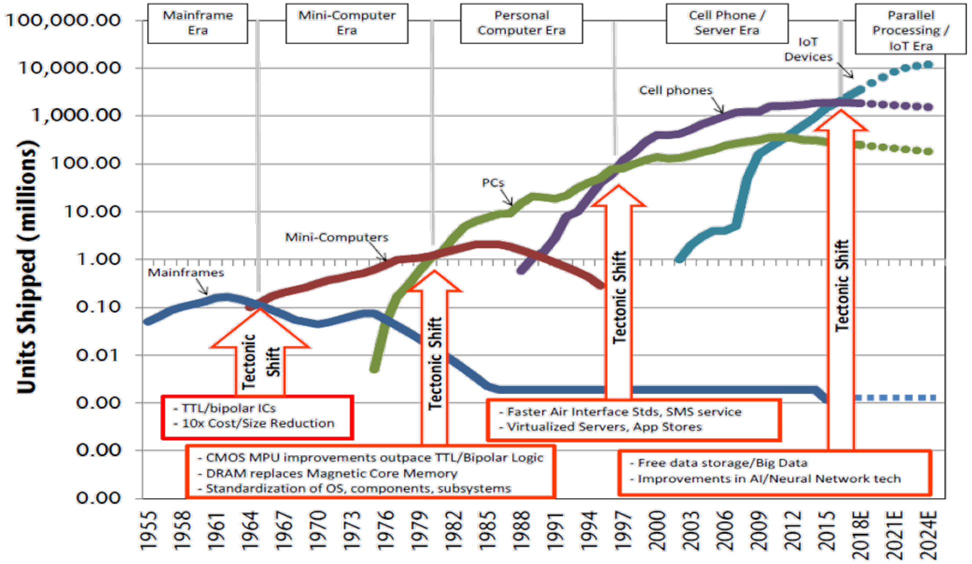
- Company that manufacturers semiconductor chips (fabricator) for clients (fabless companies) without in-house semiconductors manufacturing infrastructures

arm → tsmc →

INDUSTRY OVERVIEW – AI MEGA TREND

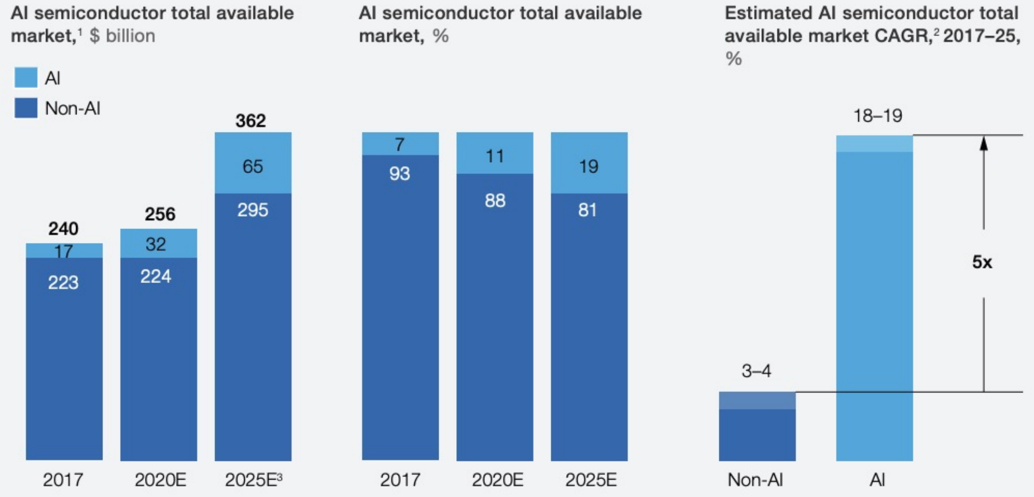
AI is bringing the next shift in tech

- AI is the 4th tectonic shift in the computing industry
- AI will disrupt and create value over many industries



AI will be the next big driver in the semiconductor industry

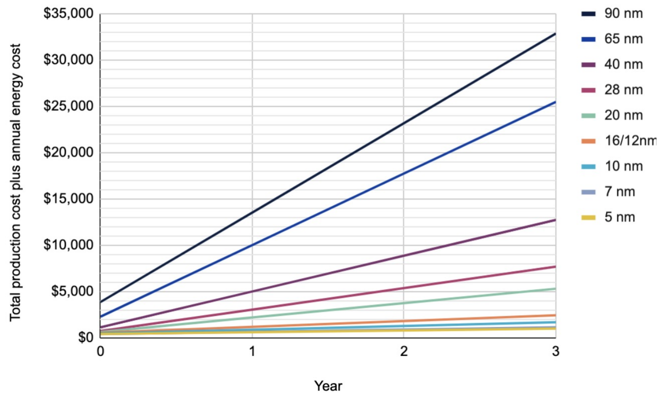
- The market has grown from $53.5B to $119.4B in 2023 and its growth is expected to be 5x the growth in the rest of the market
- Driven by generative AI and increased use of AI applications (data centers, edge infrastructure, endpoint devices)



To remain ahead of competition, AI chips need to be the fastest and most cost-effective

- Speed: the most advanced nodes provide the fastest computation
- Two cost sources: chip production cost and operating cost
- The operating cost is the dominant one, which decreases the more advanced the node is as they are more energy efficient

Cost of AI chips over time for different nodes



THESIS – TSMC is key in AI in the semiconductor industry

TSMC will capture the biggest market in AI

1. Market leadership

- **56.4%** of semiconductor foundry market share significantly ahead of competition
- Industry expected to grow to **$1T by 2030**
- Revenue mix consists of the most advanced nodes: N3(3%), N5(37%), N7(16%)

2. Leader in node manufacturing process

	Most advanced node	Expected
TSMC	2nm	2Q 2025
Intel	1.8nm	1Q 2025
Samsung	2nm	2Q 2025

- TSMC's 3nm node is expected to be on par with Intel's node in terms of PPA, leaving the 2nm far ahead
- Intel still outsources to TSMC and Samsung has issues with yield and bad performance
- TSMC is leading in GAA patent filings, key to sub 3nm nodes, with only Intel and Samsung considering GAA tech due to high costs

What separates TSMC from its peers

1. Pureplay foundry and strategic partnership

- TSMC doesn't compete with its customers
- Long-term agreements with industry leaders provide stable cash flows
- Joint Venture with Bosch, Infineon, NXP to build a Fab for auto/industrial chips in Germany

2. Financial stability and consistent CAPEX

- **Strong profitability:** 54.3% gross margin and 38.6% net income margin
- Debt-to-equity ratio of **0.3x**
- Most aggressive sales to capex, capex spend in industry

Table 2 – Capex Spend by Intel, Samsung, and TSMC ($Billions)

	2020	%YoY	2021	%YoY	2022	%YoY	2023	%YoY
Intel Logic/Foundry	12.2	-15.1%	16.7	36.5%	24.8	48.2%	30.0	21.0%
Samsung Foundry/Logic	9.4	49.3%	12.2	29.8%	11.6	-4.9%	12.4	6.4%
TSMC Foundry	17.1	15.0%	30.0	75.4%	36.0	19.9%	34.0	-6.0%

3. Superior end-to-end solution provider

- Provides advanced packaging solutions for leading backend solutions in the fabrication process
- Competitors can't capture value in this market as they have few HPC customers

How is TSMC capturing value from AI

1. Expanding into automotive industry

- Automotive industry is key driver in semiconductor industry, projected to consume 15% of global semiconductors production by 2023.
- N3 chip automotive grade variants

2. Expansion into the US

- Increasing capacity in the US with government support under CHIPS Act
- Building a $12B factory in Arizona, bringing total investment to $40B
- Long-term plan for TSMC to expand suppling chips to the US domestically

CATALYSTS

1. **Increasing demand from companies developing proprietary AI ASICs to use and migrate to 3nm**
   - E.g. Amazon Inferentia 3, Google TPU v6, Meta MTIA v2, Microsoft Athena 2

APPLE SECURES 90% OF TSMC'S 3NM CHIP PRODUCTION: WHAT DOES THIS MEAN FOR THE INDUSTRY? — Gizchina

MediaTek says it will mass produce chip with TSMC's 3nm process in 2024 — THE TIMES OF INDIA

2. **Announcements from companies adopting N2 node from 2025**
   - E.g. Apple, NVDA, AMD, AWS, QCAM, Mediatek, Broadcom

Apple, Intel likely to be first customers for TSM C's 2nm node — computing

TSMC chips away at the competition with 2nm production set for 2025 — The Register

3. **Movement to consumer tech products**
   - Consumer tech companies (PCs, smartphones) have announced the inclusion of dedicated AI chips within their devices

Google To Switch To TSMC's 3nm Process For Its Fully Custom Tensor SoC, Company Reportedly Ditching Samsung After Two Years — wccftech

Microsoft exec: AI will reinvent Windows user experience, define 'our time' — COMPUTERWORLD

TECHNICAL ANALYSIS

Moving Averages

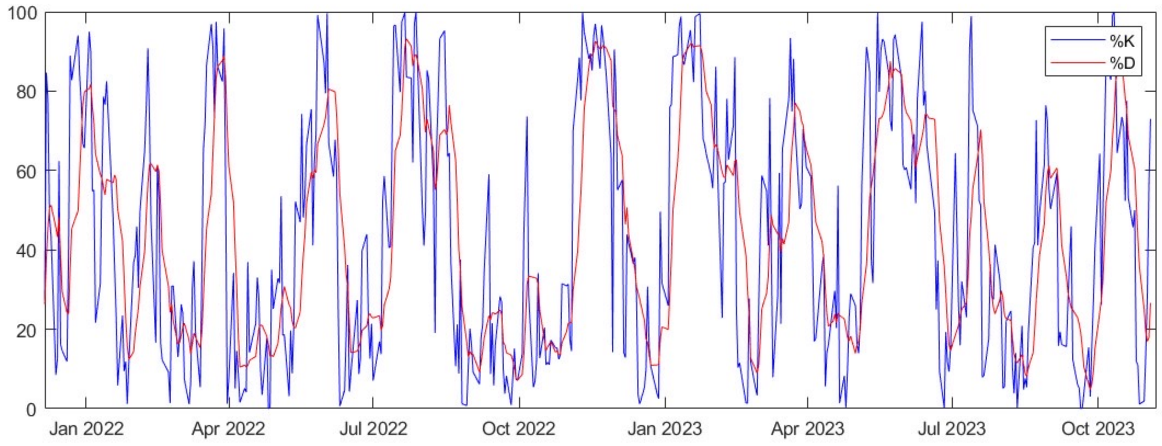
1 Year in 1 Day Interval			
Stock Price		90.48	
Moving Average	Period (D)	Value	Action
Exponential Moving Average	10	88.52	Buy
Simple Moving Average	10	88.58	Buy
Exponential Moving Average	20	88.89	Buy
Simple Moving Average	20	89.71	Buy
Exponential Moving Average	30	89.21	Buy
Simple Moving Average	30	88.53	Buy
Exponential Moving Average	50	90.11	Buy
Simple Moving Average	50	89.61	Buy
Exponential Moving Average	100	91.41	Sell
Simple Moving Average	100	94.38	Sell
Exponential Moving Average	200	91.29	Sell
Simple Moving Average	200	92.67	Sell
Volume Weighted Moving Average	20	89.8	Buy
Hull Moving Average	9	87.34	Buy
Moving averages			Buy

Change from a downtrend to an uptrend:

- The stock price has crossed above the short-term MAs, remaining under the long-term MAs
- 20- and 50-day MAs show an uptrend
- If the 100-day MA is crossed, it will be a strong buy signal

TECHNICAL ANALYSIS

Oscillators



RSI at 54.22

Neutral signal

MACD at -0.44

Neutral-Bullish signal

Stochastic %K at 72.9
%D (7 day) at 26.6
Oversold signal

VALUATION – DCF (Base Case)

Assumption	2024F	2025F	2026F	2027F	2028F	2029F	2030F	2031F	2032F	2033F	Terminal Value
EBIT	1,110,804	1,281,583	1,478,183	1,704,459	1,960,828	2,255,298	2,593,476	2,981,788	3,427,597	3,886,780	
EBIT (1-T)	987,228	1,139,008	1,313,736	1,514,839	1,742,688	2,004,398	2,304,954	2,650,067	3,046,279	3,454,379	
D&A	551,433	638,774	737,815	850,125	977,480	1,121,896	1,285,659	1,471,360	1,681,940	1,920,730	
Chg. In WC	14,027	14,313	14,454	14,406	14,123	13,547	12,610	11,231	9,317	27,278	
Capex	(1,213,535)	(1,376,105)	(1,560,455)	(1,769,501)	(2,006,553)	(2,275,362)	(2,580,182)	(2,925,839)	(3,317,804)	(3,762,279)	
Net FCF	**339,153**	**415,990**	**505,551**	**609,869**	**727,738**	**864,480**	**1,023,040**	**1,206,819**	**1,419,732**	**1,640,108**	
Terminal Value											**28,237,359**
Discount Factor	0.9	0.8	0.8	0.7	0.7	0.6	0.5	0.5	0.5	0.4	0.4
PV of FCFF	**311,200**	**350,243**	**390,566**	**432,324**	**473,359**	**515,957**	**560,266**	**606,438**	**654,627**	**693,910**	**11,946,888**
Firm Value (TWD)	**16,935,777**										
Less: Debt	(1,041,651)										
Less: Minority Interest	(16,832)										
Plus: Cash	1,296,065										
Equity Value (TWD)	**17,173,359**										
Equity Value (USD)	**532,374**										
Shares Outstanding (USD)	5,186										
Price per Share (USD)	**103**										
Current Share Price (USD)	90										
Potential Upside	14%										

VALUATION – Trading Comparables

We compared TSMC to 5 other semiconductor companies with Revenue >20B, Listed in the US

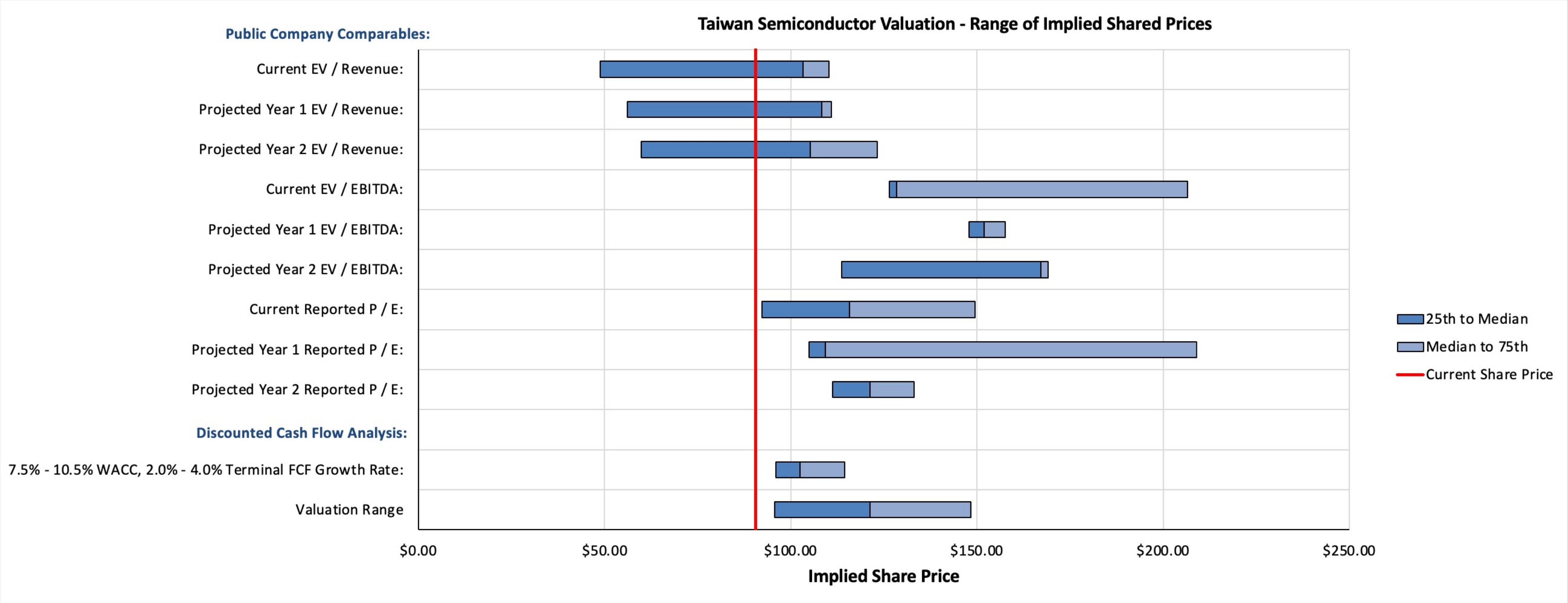
- Intel
- Qualcomm
- Broadcom
- Texas Instruments
- Advanced Micro Devices (AMD)

We calculated the semiconductor industrial multiples with data and projections of these 5 companies

Data and Projection Source: Capital IQ

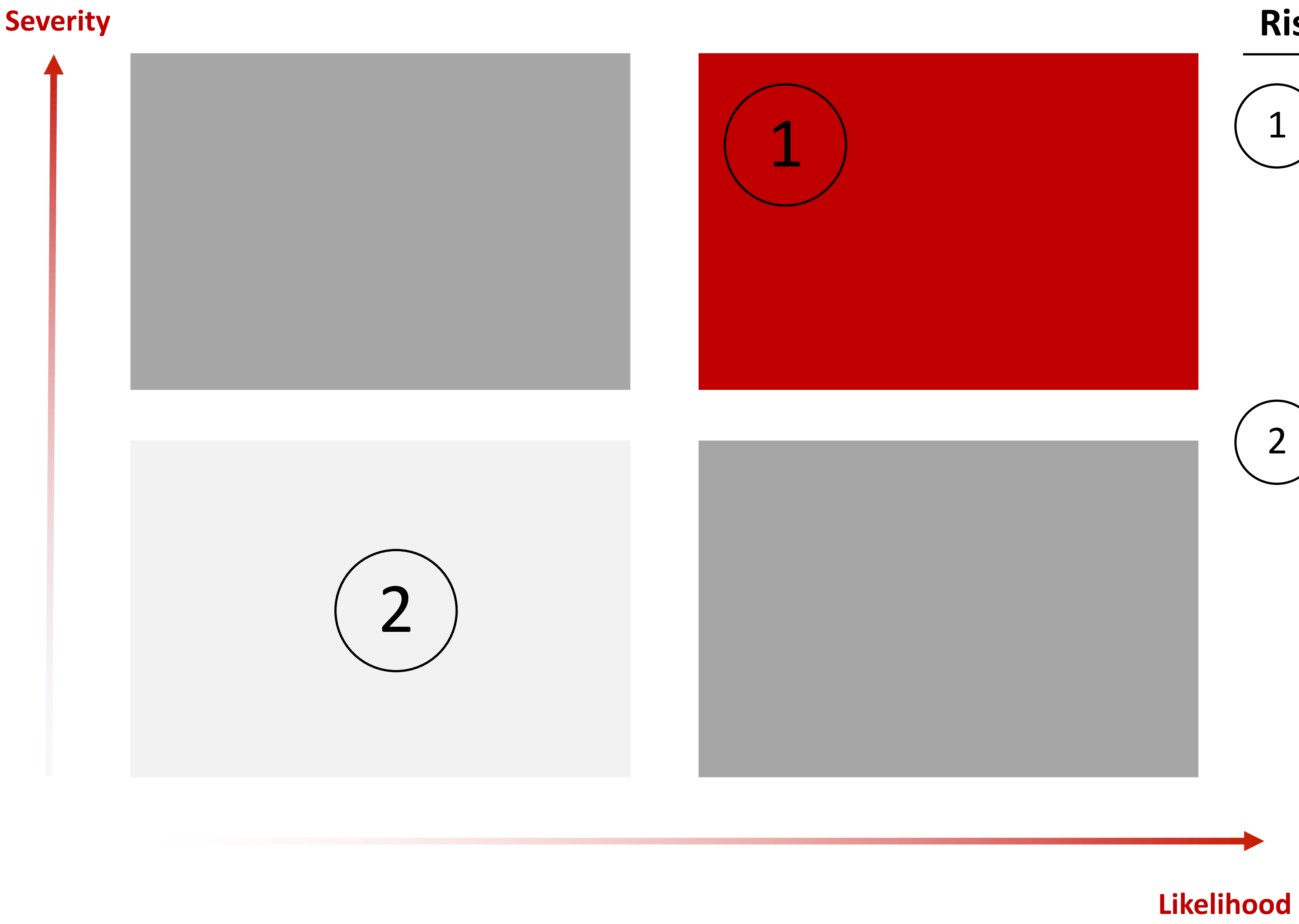
Methodology Name	Maximum Multiple	75th Percentile Multiple	Median Multiple	25th Percentile Multiple	Minimum Multiple	Applicable Company Figure	Minimum Multiple	25th Percentile Multiple	Median Multiple	75th Percentile Multiple	Maximum Multiple
Public Company Comparables:											
2022 EV / Revenue:	11.7 x	7.3 x	6.8 x	3.1 x	3.0 x	$ 76,074.5	$ 47.10	$ 48.71	$ 103.32	$ 110.16	$ 174.59
FY1 EV / Revenue:	10.8 x	7.7 x	7.6 x	3.8 x	3.5 x	71,768.1	51.69	56.10	108.35	110.88	153.22
FY2 EV / Revenue:	10.0 x	7.6 x	6.5 x	3.6 x	3.1 x	81,382.4	51.82	59.88	105.25	123.24	160.23
2022 EV / EBITDA:	36.5 x	20.1 x	12.3 x	12.1 x	8.2 x	52,398.3	86.46	126.37	128.42	206.52	371.98
FY1 EV / EBITDA:	45.0 x	16.9 x	16.3 x	15.9 x	13.0 x	47,091.4	121.50	147.76	151.85	157.54	412.01
FY2 EV / EBITDA:	27.3 x	15.7 x	15.6 x	10.5 x	9.9 x	54,503.6	107.64	113.56	167.10	169.02	290.58
2022 P / E:	31.3 x	22.7 x	17.6 x	14.0 x	10.1 x	34,158.9	66.69	92.21	115.68	149.49	206.05
FY1 P / E:	40.4 x	39.1 x	20.4 x	19.6 x	18.1 x	27,713.8	96.78	104.94	109.26	209.03	216.00
FY2 P / E:	28.3 x	21.6 x	19.7 x	18.0 x	12.7 x	31,989.0	78.14	111.19	121.30	133.15	174.41
DCF 7.5% - 10.5% WACC, 2.0% - 4.0% Terminal FCF Growth Rate:							87.13	95.91	102.45	114.53	154.49
Valuation Range (Average of the quartiles)							79.49	95.66	121.30	148.35	231.36

VALUATION – Football Field



Valuation Range: $95.66 – 148.35

INTERNAL RISK ANALYSIS

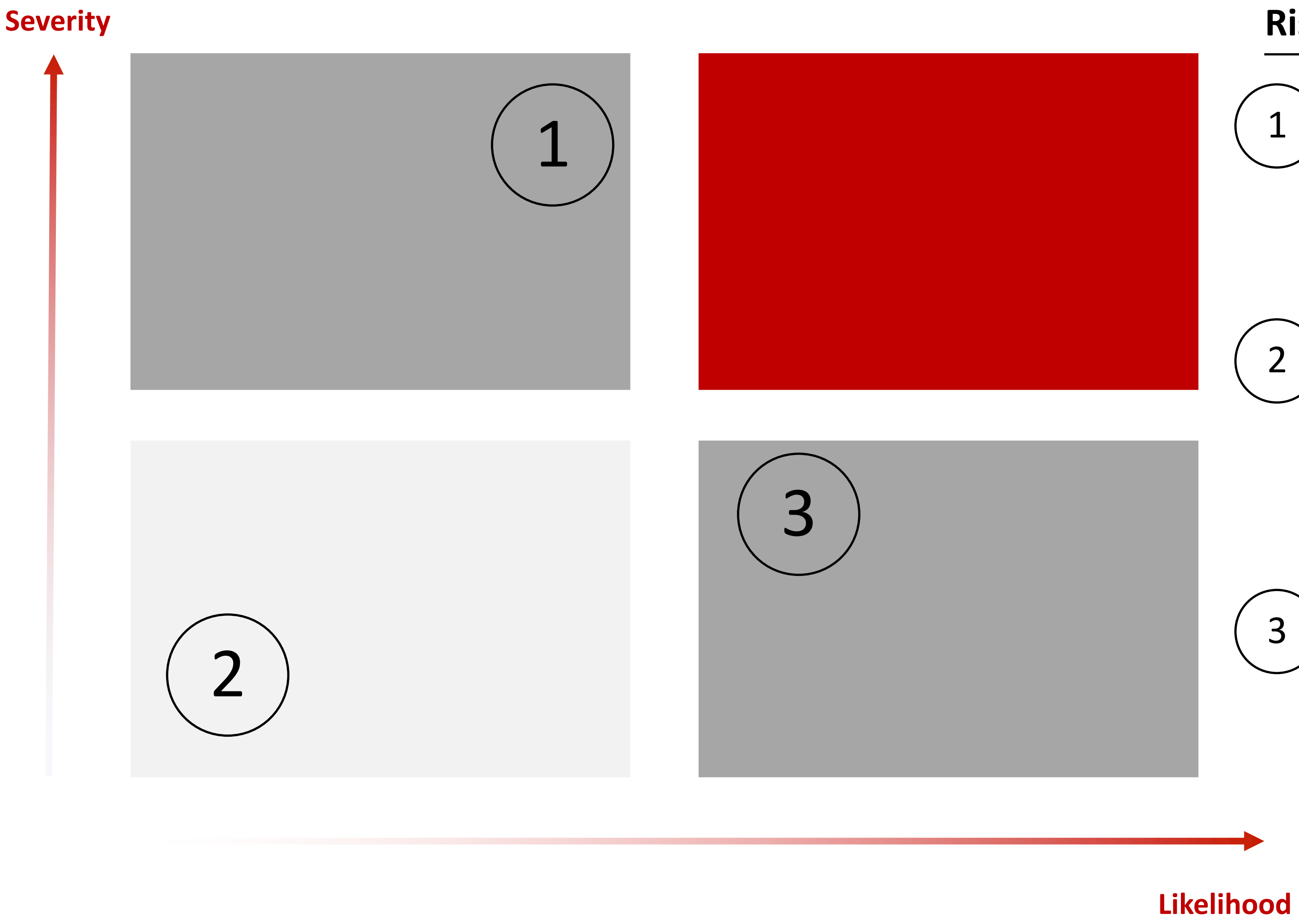


Risk Description

1. **High reliance on few key customers**
   - Apple accounted for 23% of TSMC's revenue in 2022
   - Makes TSMC vulnerable to demand shifts, changes in customer preferences

2. **Huge investments in R&D**
   - TSMC spent $5.5B in R&D in 2022
   - Essential to maintain competitive edge
   - Can pressure short-term profitability and cash flow

EXTERNAL RISK ANALYSIS



Risk Description

1 **Taiwan-China geopolitical tension**
- China is TSMC’s key market
- TSMC is expanding manufacturing facilities to Germany and USA

2 **Cyclical industry**
- Semiconductor industry is characterized by periods of overcapacity followed by shortages

3 **USA-China geopolitical tension**
- TSMC’s key clients belong on both sides (Apple, Huawei)
- Border trade tensions

Questions

Appendix

APPENDIX 1 – FINANCIAL FORECAST

Profit and Loss					
Y/E Dec (TWD millions)	21A	22A	23F	24F	25F
Revenues	1,587,415	2,263,891	2,135,737	2,421,849	2,746,291
Cost of Revenues	(767,878)	(915,536)	(966,170)	(1,078,649)	(1,203,926)
Gross Profit/Loss	**819,537**	**1,348,355**	**1,169,567**	**1,343,200**	**1,542,365**
SGA Expense	(44,488)	(63,445)	(59,854)	(65,450)	(71,472)
Operating Profit/Loss	**650,108**	**1,122,061**	**962,490**	**1,110,804**	**1,281,583**
EBITDA	**1,072,503**	**1,559,315**	**1,436,901**	**1,662,238**	**1,920,357**
Finance Income	5,709	22,422	21,642	22,898	24,311
Finance Expenses	(5,414)	(11,750)	(5,825)	(6,536)	(7,213)
Pre-tax Profit/Loss	**663,126**	**1,144,191**	**993,343**	**1,143,523**	**1,316,528**
Income Tax	(66,053)	(127,290)	(110,509)	(127,216)	(146,463)
Profit/Loss for Period	**597,073**	**1,016,901**	**882,835**	**1,016,307**	**1,170,065**
Minority Interest	533	370	321	370	426
Net Profit	**596,540**	**1,016,530**	**882,513**	**1,015,937**	**1,169,639**

Balance Sheet					
Y/E Dec (TWD millions)	21A	22A	23F	24F	25F
Cash & equivalents	1,064,990	1,342,814	1,296,065	1,371,288	1,455,932
Receivables	197,586	231,340	225,026	255,171	289,355
Others	221,204	261,144	282,963	311,249	342,753
Total Current Assets	**1,607,073**	**2,052,897**	**2,021,653**	**2,155,307**	**2,305,638**
Net Fixed Asset	2,007,853	2,735,751	3,331,511	3,993,612	4,730,943
Other Non-Current Asset	26,822	25,999	26,779	27,583	28,410
Total Assets	**3,725,503**	**4,964,779**	**5,530,074**	**6,326,633**	**7,215,123**
ST. Bank loan	121,664	21,917	25,086	28,148	31,066
Payables	431,643	637,886	673,164	745,622	825,623
Other current Liabilities	181,630	265,109	265,109	265,109	265,109
Total Current Liabilities	**739,503**	**944,227**	**985,465**	**1,063,683**	**1,149,174**
LT.Debt	634,144	868,861	994,459	1,115,843	1,231,527
Other LT Liabilities	181,123	191,203	191,203	191,203	191,203
Total Liabilities	**1,554,770**	**2,004,290**	**2,171,127**	**2,370,729**	**2,571,903**
Minority Interest	2,447	14,836	16,832	19,824	23,268
Total Equity	**2,170,733**	**2,960,489**	**3,358,947**	**3,955,904**	**4,643,220**

Cash Flow					
Y/E Dec (TWD millions)	21A	22A	23F	24F	25F
Net Profit	596,540	1,016,530	882,513	1,015,937	1,169,639
D&A	390,157	406,488	474,411	551,433	638,774
Changes in Working Capital	(18,360)	121,723	19,773	14,027	14,313
Operating Cash Flow	**1,116,817**	**1,619,481**	**1,376,698**	**1,581,397**	**1,822,727**
Capital Expenditure	(814,693)	(1,134,386)	(1,070,170)	(1,213,535)	(1,376,105)
Others	(30,163)	(77,643)	(780)	(803)	(827)
Investing Cash Flow	**(844,856)**	**(1,212,028)**	**(1,070,950)**	**(1,214,338)**	**(1,376,933)**
Net - Borrowing	388,155	149,717	131,559	127,144	121,173
Other Financing	(247,713)	(337,742)	(484,055)	(418,980)	(482,323)
Financing Cash Flow	**140,442**	**(188,025)**	**(352,496)**	**(291,836)**	**(361,150)**
Net - Cash Flow	404,820	277,824	(46,749)	75,223	84,644
Cash at beginning	660,171	1,064,990	1,342,814	1,296,065	1,371,288
Cash at ending	1,064,990	1,342,814	1,296,065	1,371,288	1,455,932

Key Ratios					
Y/E Dec (TWD millions)	21A	22A	23F	24F	25F
Gross Profit Margin (%)	51.6	59.6	54.8	55.5	56.2
Operating Margin (%)	41.0	49.6	45.1	45.9	46.7
EBITDA Margin (%)	67.6	68.9	67.3	68.6	69.9
Net Profit Margin (%)	37.6	44.9	41.3	41.9	42.6
Revenue Growth	18.5	42.6	(5.7)	13.4	13.4
EPS Growth	19.4	45.4	(7.9)	15.7	15.5
Net Gearing (%)	(0.1)	(0.1)	(0.1)	(0.1)	(0.0)

Major Assumptions					
	21A	22A	23F	24F	25F
Wafer Shipment (kpcs)	14,179	15,253	10,525	11,261	12,050
ASP (TWD/pcs)	111,955	148,423	173,655	184,074	195,118